

TED[15] 5241
(REVISION-2015)
Signature.....

Reg. No.....

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/
MANAGEMENT/
COMMERCIAL PRACTICE

THERAPUTIC EQUIPMENT

[Time : 3 hours]

(Maximum marks : 100)

PART - A

(Maximum marks: 10)

Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define fibrillation?
2. Define ultra sound?
3. Name the cycling mechanisms used in ventilators?
4. List any two application of syringe pump?
5. Draw the Lawn waveform?

(5x2=10)

PART - B

(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Describe Automatic Implantable Defibrillators?
2. List the design requirements of cardiac pacemakers?
3. Explain the different cycling mechanism in ventilators?
4. List the applications of radio therapy?
5. Explain modes of operation of ventilators?
6. Describe pumps used in heart lung machine?
7. Explain the principle and operation of endoscopy?

(5x6=30)

PART - C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT – I

III. a) Describe programmable pacemakers with block diagram? (9)

b) Explain the principle of defibrillators? (6)

OR

IV. a) Explain the types of pain relief through electrical stimulation? (9)

b) Give an account on power sources of cardiac pacemakers? (6)

UNIT II

V.a) Describe the working of ultra sonic diathermy machine? (8)

b) List the application of laser in medicine? (7)

OR

VI.a) Explain the basic operation of He-Ne laser with diagram? (8)

b) Elaborate the principle and operation of ultra sonic lithotripter? (7)

UNIT III

VII. a) Describe the block diagram of anesthesia machine? (10)

b) Explain principle and use of oxygenators? (5)

OR

VIII. a) Explain the operation of the block diagram of heart lung machine? (10)

b) Explain the working of bedside monitors? (5)

IX. a) Describe the principle and operation of LINACC? (8)

b) Explain the working of syringe pump? (7)

OR

X. a) Explain different types of endoscopy with applications? (8)

b) Describe the biological effect of radio therapy? (7)

(15*4=60)